

DRAFT | Hybrid recommendation: three-split frozen evaluation

DSAIT4335 Recommender Systems

Group: 24

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Frozen held-out means across 3 overlapping splits; contributors pending

Every primary table averages separately frozen evaluations for seeds 2026, 2027, 2028. Model identities and family roles are aligned from frozen metadata; no test-based selection occurs.

One page is allocated to each major task, with at most 200 discussion words per task. The report content JSON contains the complete aggregate metrics without individual histories or recommendations.

Contributor metadata is incomplete; remaining names and the individual peer-feedback workbook must be completed before submission. AI assistance was used for implementation, analysis and report preparation.

This revision consolidates the three required tasks. The original seven-page report and its field/rejection appendices are preserved separately; the new grouping appendix is labelled by its own evidence stage.

Method references

- [1] Steck (2019). Embarrassingly Shallow Autoencoders for Sparse Data. WWW.
<https://arxiv.org/abs/1905.03375>
- [2] Rendle et al. (2009). BPR: Bayesian Personalized Ranking from Implicit Feedback. UAI.
<https://arxiv.org/abs/1205.2618>
- [3] Steck (2018). Calibrated Recommendations. RecSys. doi:10.1145/3240323.3240372. Our JSD implementation is an adaptation, not an exact reproduction.
- [4] Ning and Karypis (2011). SLIM: Sparse Linear Methods for Top-N Recommender Systems. ICDM. doi:10.1109/ICDM.2011.134.
- [5] Kabbur, Ning and Karypis (2013). FISM: Factored Item Similarity Models for Top-N Recommender Systems. KDD. doi:10.1145/2487575.2487589.
- [6] He et al. (2020). LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. <https://arxiv.org/abs/2002.02126>
- [7] He et al. (2017). Neural Collaborative Filtering. <https://arxiv.org/abs/1708.05031>
- [8] Wang et al. (2019). Neural Graph Collaborative Filtering. <https://arxiv.org/abs/1905.08108>
- [9] Aioli (2013). Efficient Top-N Recommendation for Very Large Scale Binary Rated Datasets. RecSys. doi:10.1145/2507157.2507189. Cosine KNN uses the course adapter, not all paper variants.
- [10] Harper and Konstan (2015). The MovieLens Datasets: History and Context. ACM TiiS. Rating timestamps are recording times, not consumption times.
<https://files.grouplens.org/papers/harper-tiis2015.pdf>

DRAFT | Task 1: individual models and hybrids

MovieLens 100K, full-catalog ranking at $k=10$; 3 overlapping random splits (2026, 2027, 2028), each 943 users; held-out interactions by seed: 9596/9596/9596. All observed ratings are relevant. Inference uses frozen TRAIN fits and masks TRAIN+validation history. Tables average per-split macro metrics, not pooled users or independent dataset replications.

Meta-fit users select expert settings and fit standardized-score coefficients; development users choose hybrid penalties before freezing. User fusion adds score-by-activity/genre-entropy interactions; item fusion adds score-by-popularity interactions; disagreement adds cross-expert score deviation; context combines these. Pairwise variants fit positive-minus-negative feature differences. RRF sums reciprocal ranks with offset 60; group-switch selects a meta-fit expert per activity tercile.

Static ridge is unconstrained. The lecture sum-to-one baseline allows negative weights; calibrated ridge first learns an affine response alignment per expert on meta-fit users. Effective raw-z coefficients need not sum to one, and scores are unbounded.

Mean held-out nDCG@10; individual models and frozen hybrid families

Model	nDCG	Model	nDCG
BPR	0.2847	SignedChannelsLinear	0.2870
ContrastTransfer	0.1599	UserKNN	0.2805
EASE	0.3215	calibrated	0.3339
ExactPop	0.1412	constrained	0.2677
FISMCorrected	0.2717	context	0.3326
GenreContent	0.0232	context-pairwise	0.3100
ItemKNN	0.2809	disagreement	0.3370
LightGCN	0.3003	item	0.3345
NGCF	0.2863	static	0.3356
NeuMF	0.2678	static-pairwise	0.3290
PositiveEASE	0.2847	user	0.3340
Random	0.0089	rrf	0.3205
SLIMElastic	0.3134	group-switch	0.3211

Discussion

Disagreement-conditioned fusion scores 0.3370 versus 0.3161 for the validation-selected standalone reference (+6.6% relative); static fusion scores 0.3356. The reference was selected separately in each split; these are descriptive held-out differences. Response alignment raises constrained nDCG from 0.268 to 0.334: raw sum-to-one restrictions also impose a response-scale cost. Negative weights remain allowed and are not probabilities. FISMCorrected jointly repairs the supplied logit/loss mismatch and target-history inclusion, retaining its non-squared regularizer; it is not an exact paper reproduction. LightGCN received a declared longer training budget following improving validation curves. Test metrics never choose models or budgets.

DRAFT | Task 2: effectiveness and interpretation

Metrics are independent of RecBole. Accuracy macro-averages users with held-out positives. Activity terciles and the top-20% popularity head use TRAIN only. Genre diversity is pairwise Jaccard distance. Head/tail recall conditions on users with positives in that group; exposure is recommendation-slot share. All rows are equal-weight means of three frozen splits. Coefficients are frozen static-regression weights.

Mean static-regression coefficients on standardized expert scores

Model	Weight	Model	Weight
BPR	+0.004	LightGCN	+0.001
ContrastTransfer	+0.006	NGCF	+0.008
EASE	+0.033	NeuMF	+0.050
ExactPop	+0.001	PositiveEASE	+0.018
FISMCorrected	+0.011	SLIMElastic	+0.029
GenreContent	+0.006	SignedChannelsLinear	+0.002
ItemKNN	+0.015	UserKNN	-0.004

User-group nDCG; conditional head/tail item recall

Model	Sparse	Medium	Dense	Head	Tail
EASE	0.254	0.260	0.450	0.383	0.010
SLIMElastic	0.251	0.254	0.434	0.361	0.029
context	0.259	0.273	0.466	0.387	0.016

User-group genre diversity; head/tail item exposure

Model	Sparse	Medium	Dense	Head	Tail
EASE	0.801	0.807	0.823	0.985	0.015
SLIMElastic	0.802	0.816	0.828	0.956	0.044
context	0.789	0.794	0.810	0.981	0.019

Discussion

Coefficients reflect correlated score associations, not causal credit; NeuMF's largest weight does not make it the strongest individual model. SLIM allocates more slots to the tail and has higher tail recall than EASE, despite lower overall nDCG. Context fusion improves user-group accuracy while reducing genre diversity in every group. These differences motivate explicit accuracy/diversity and head/tail trade-offs, rather than declaring one model universally better. Our earlier custom field rose from nDCG 0.014 under conditional-rating training to 0.268 under joint-record training at a 400-epoch cap, still below EASE; adaptive routing had no consistent demonstrated advantage. Its full controls remain in the archived report. A separate TRAIN-only audit finds 70.14% of records in same-user timestamp ties and pair-weighted genre Jaccard 0.234 versus 0.184 after within-user shuffling. This establishes recording-group structure, not predictive benefit or viewing sessions [10]. It motivates the separately declared grouping study while keeping hidden target timestamps unavailable. Reused development evidence is exploratory; no new method or superior hybrid is inferred from coherence alone.

DRAFT | Task 3: societal aspects and reranking

The fixed-strength comparison uses 0.5 for diversity, calibration [3] and exposure rerankers. Genre JSD measures deviation from historical genre shares; lower values mean better calibration. Head exposure measures allocation to popular items. RRF ordering comparisons retain full ranking support; objective values accompany accuracy.

UPD is per-user popularity-profile JSD. The base contextual hybrid's mean discounted catalog exposure Gini is 0.927; lower values mean less concentrated allocation.

Mean held-out accuracy and societal trade-offs

Reranker	nDCG	Diversity	JSD	Head	UPD
Base context	0.333	0.798	0.141	0.981	0.149
Diversity	0.307	0.895	0.131	0.967	0.133
Calibration	0.329	0.806	0.125	0.980	0.148
Item exposure	0.318	0.793	0.141	0.931	0.097
Popularity calibration	0.333	0.795	0.141	0.961	0.124

Before/after RRF: nDCG and objective value (diversity, JSD, head share or UPD)

Objective	Bef. nDCG	Aft. nDCG	Bef. goal	Aft. goal
diversity	0.315	0.298	0.832	0.887
calibration	0.321	0.313	0.142	0.132
exposure	0.321	0.304	0.979	0.946
popularity_calibration	0.320	0.320	0.154	0.142

Group effects: utility-policy users and fixed-exposure item groups

Group	Base	After	Metric
Sparse users	0.259	0.260	nDCG
Medium users	0.273	0.273	nDCG
Dense users	0.466	0.465	nDCG
Head items	0.387	0.376	Recall
Tail items	0.016	0.039	Recall

Discussion

Diversity rises from 0.798 to 0.895, while nDCG falls from 0.333 to 0.307. Exposure reranking raises tail recall from 0.016 to 0.039. Across seeds and activity groups, policy utility retention ranged from 99.7% to 101.0%. The independent calibration target is not a held-out guarantee. Smaller gaps can hide losses for every group, so absolute utility accompanies disparity. Catalog-proportional exposure is a declared objective, not a universal fairness definition. With H head and T tail candidates, feasible head counts satisfy $\max(0, k-T) \leq \text{heads} \leq \min(k, H)$. Reranking before fusion changes information supplied to the fusion rule, so its order can change both accuracy and the target objective. Every frozen strength remains in aggregate evidence; test outcomes never retune strength.

Appendix: do rating-recording groups improve ranking?

REVIEW DRAFT | Exploratory nested split of original TRAIN; original validation/test unopened.

Motivation: 70.14% of original TRAIN ratings share a user's timestamp. Within-group genre Jaccard is 0.234 versus 0.184 across 100 within-user shuffles. This is descriptive evidence.

Known constrained ridge, singleton + pair features; own-target features excluded. F/D/A = 65,518/7,645/7,645 records; 943 users. All 39 candidates use F context; D selects, A assesses after sealing. All 1,682 items eligible except F observations; no refit or target timestamps.

All-recorded assessment at ten (every selected arm)

Model	lambda / beta	nDCG	Recall
Tuned EASE	300 / 0	0.1730	0.2043
Unordered pairs	250 / 0.1	0.1727	0.2027
Recording groups	250 / 0.1	0.1726	0.2035
Shuffle 1	250 / 0.1	0.1728	0.2031
Shuffle 2	250 / 0.1	0.1730	0.2039
Shuffle 3	250 / 0.1	0.1726	0.2035

Recording groups minus control: paired nDCG

Control	Mean change	95% interval
vs EASE	-0.0004	[-0.0020, +0.0013]
vs unordered	-0.0002	[-0.0014, +0.0012]
vs shuffle mean	-0.0003	[-0.0012, +0.0006]

Intervals: 2,000 paired-user resamples, descriptive and not multiplicity adjusted. Shuffle comparison averages metrics, never scores. Full subgroup/secondary metrics remain in aggregate evidence.

Discussion

Recording groups did not meet the declared research-priority rule: at least 10% relative nDCG gain over tuned EASE and higher means than unordered pairs and shuffled groups. The measured EASE-relative change was -0.23%. Coherent recording groups alone do not establish predictive value. The unordered control tests the grouping restriction; timestamp shuffles retain each user's movies and group sizes. They do not identify exposure, interface design, or watching sessions. Pairwise linear recommendation is established; the contribution here is a controlled input-structure test. This single nested split follows earlier dataset exploration. Its assessment is separated from model selection, but is not fresh population confirmation or directly comparable with the preceding test tables.

Prior: Steck & Liang (2021), Higher-Order Interactions in Linear CF. doi:10.1145/3460231.3474273. Protocol, source hashes and all candidates accompany this report.